AGOG OPEN CALL 2026 · CLIMATE FUTURES + IMMERSIVE MEDIA

Lake Champlain Portal

A browser-based WebXR experience that lets you descend through 15,000 years of Lake Champlain's past, present, and future, *as the land sinks, the sea rises, and the valley remembers.*

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The ground beneath your feet has never been still

Buried

A mile of glacial ice covered the valley, depressing the land beneath it.

1

Flooded

As the ice collapsed, a saltwater sea rushed in from the Atlantic. Beluga whales swam where Burlington now stands.

2

Rebounding

As the land slowly rose and the sea retreated, freshwater reclaimed the valley, becoming Lake Champlain.

3

Transforming again

A warming climate is now compressing changes that once took millennia into decades.

4

The people who first inhabited this valley arrived as that transformation was still underway.

Grounded in real geological data

Every temporal state the viewer experiences corresponds to a documented paleogeographic reconstruction.

Vermont Geological Survey

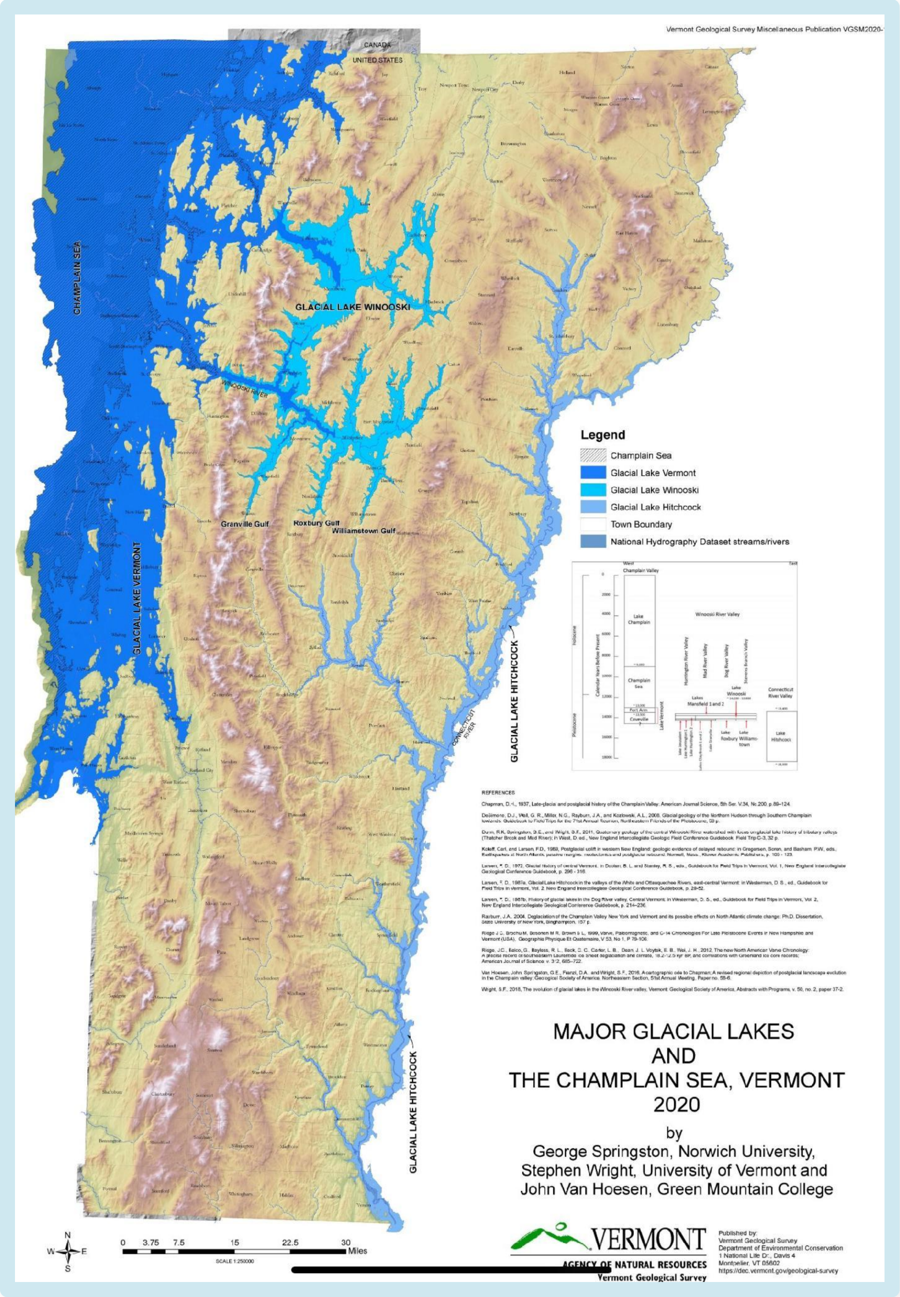
Published mapping of the Champlain Sea extent, Glacial Lake Vermont, and Glacial Lake Winooski (Springston, Wright & Van Hoesen, 2020) provides the paleoshoreline record this project animates.

Glacial isostatic adjustment models

Published GIA datasets (e.g. ICE-6G) provide crustal depression estimates through the last glacial maximum, the foundation for the terrain-sinking sequence.

Bathymetric and underwater survey data

NOAA and Lake Champlain Maritime Museum survey data will inform the present-day lakebed and the underwater layer of the descent.



The Experience

Step onto the shore. Then descend, through water, through time.

Present shoreline

Stereoscopic source footage of the lake today: a weathered dock, dead trees, submerged logs. In passthrough AR, this overlays the viewer's real room.

The descent begins

The viewer sinks through the water column. The land falls away beneath them, animated from real elevation data and isostatic depression models.

The sea rushes in

Salt water from the Champlain Sea floods the valley from the north, rendered as VR180 stereo video built from AI-reconstructed scenes.

Surface, transformed

The viewer returns to the present, and can move forward into projected climate futures. The room is the same. The viewer is not.

Built on a working prototype

- WebXR + Three.js, runs in any modern browser, no install
- Passthrough AR with VR fallback (Immersive Web SDK)
- Compositor-level stereo video via XRQuadLayer
- High-res DEM terrain data of the Champlain basin
- On-site stereoscopic source footage + spatial audio
- AI-generated 3D assets refined via Blender MCP pipeline
- XR emulation (IWER) for access without a headset

From the working prototype

Live render of the Champlain shoreline.

From footage to deep time: the production pipeline

A scalable, AI-assisted approach to place-based immersive reconstruction

Source footage

On-site stereoscopic (SBS) video of the present-day Champlain shoreline, captured at 90fps.

AI 3D reconstruction

“image-blaster” (Claude Code skill) + Hunyuan 3D generate PBR-textured meshes of environmental objects from video frames.

Blender MCP refinement

Generated meshes are placed into scientifically grounded scenes representing past and future states, refined via natural-language Blender control.

VR180 stereo render

Each temporal state is rendered as stereoscopic video and delivered through the WebXR compositor layer already built into the prototype.

AI is not used to replace creative judgment. It makes an otherwise prohibitively labor-intensive documentary reconstruction pipeline tractable for a solo developer, in service of a story grounded in real data and a real place.

Who this is for

Anyone ready to feel, not just understand, that they are living at a hinge point in the planet's history.

Primary: the Champlain watershed community

Vermont and New York residents, students in regional schools, and visitors to the Lake Champlain Maritime Museum and ECHO Leahy Center, people with a geographic and civic stake in this place who may not yet have a visceral relationship to its history or climate vulnerability.

Secondary: a global audience

Browser-based, no-installation WebXR is the only immersive distribution model that scales worldwide without hardware barriers. The story of glacial rebound and climate transformation is not Vermont's alone, it is Earth's story told through one valley. The temporal descent framework is designed to be adaptable to other watersheds and communities.

Whose voices are centered

11,300+ years of Abenaki presence in the Champlain Valley

The Abenaki arrived as the Champlain Sea was still receding, and carry place-based knowledge of this landscape that predates written geological records. The project centers Abenaki voices as essential, not supplementary, to the experience's narrative layer.

Funded, not an afterthought

Community engagement with Abenaki knowledge holders is a dedicated, funded phase of the work plan.

Voice, not narration

The subtitle layer is designed to carry community voice, developed through dialogue rather than imposed by the development team.

Consent before content

No Abenaki cultural knowledge will be incorporated without formal permission and ongoing relationship as principal stakeholders.

Secondary stakeholders: lakeside agricultural, fishing, and recreational communities of Vermont and New York, whose present-day relationship to the lake anchors the experience in contemporary stakes.

Partnerships and distribution

INSTITUTIONAL PARTNER

Lake Champlain Maritime Museum

Anticipated partner. Underwater Archaeology program (systematic surveys since 1996, Vermont Underwater Historic Preserve System) and existing virtual shipwreck tours, bringing bathymetric and survey data into the descent experience. Developer already connected.

DATA + COMMUNITY

Lake Champlain Basin Program

Watershed data publications inform the scientific foundation of the project. Its binational Vermont / New York community network is a distribution pathway to people closest to the lake.

TO BE ESTABLISHED

Abenaki knowledge holders & tribal organizations

Formal partnership to be built through the funded work, centering community voice in the narrative layer with consent at every stage.

SCIENTIFIC VALIDATION

VT Agency of Natural Resources & VT Geological Survey

Data access and scientific validation for terrain depression, isostatic adjustment, and paleoshoreline modeling.

INSTALLATION VENUE

ECHO Leahy Center for Lake Champlain

Outreach for a Burlington installation as part of the public deployment plan.

TECHNICAL NETWORK

WebXR developer community

W3C Immersive Web Working Group and broader ecosystem already inform the technical architecture, and the open-source, browser-based design creates natural distribution through developer and education networks.

12-month path to launch

Video production, terrain integration, and community narrative development run in parallel to hit a public launch tied to LCMM's Abenaki Heritage Weekend.

Months 1–2	**Data acquisition**	Glacial isostatic adjustment models, bathymetric data (NOAA / LCBP), paleoshoreline reconstructions. Community outreach begins with Abenaki organizations and LCMM.
Months 3–5	**Terrain integration**	DEM + GIA data into the WebXR renderer. Animated depression and flooding sequence. Temporal navigation prototype. Narrative dialogue begins in parallel.
Months 6–8	**Video production**	VR180 stereo rendering for each temporal state via the AI-assisted Blender pipeline. Spatial audio production. Narrative layer development continues with Abenaki knowledge holders.
Month 9	**Soft launch**	Browser-based release for the watershed community, with subtitle content and audio narration fully integrated.
Months 10–11	**Installation prep**	Physical installation design and logistics with LCMM for the public launch event.
Month 12	**Public launch**	In-person launch event at the Lake Champlain Maritime Museum, timed to Abenaki Heritage Weekend.

The ask

$50,000

Funding request

$75,000

Overall project budget

12 months

To public launch (Abenaki Heritage Weekend)

Beyond funding, the most valuable support would be:

Community connections

Introductions to Abenaki cultural organizations and knowledge holders in the Champlain Valley.

Scientific validation

Connections to climate scientists and glaciologists working on Champlain Valley paleography and isostatic adjustment.

Distribution partners

Immersive media festivals and environmental film programs to extend reach beyond the watershed.

This is a place that has always been in motion.

The portal makes that motion visible.

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